Fiscal Year:	FY 2014	**Task Last Updated:**	FY 12/24/2013
PI Name:	Macias, Brandon Ph.D.		
Project Title:	Altered Ocular Structure/Function and Mitigation of Peri-Optic Nerve Edema During Simulated Microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-2026
Organization Name:	NASA Johnson Space Center		
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Comments:	NOTE: Became civil servant fall 2020; previously KBR/NASA Johnson Space Center. Prior to that until 2016, was at the University of California, San Diego.		
Project Type:	Ground	**Solicitation / Funding Source:**	2012 NSBRI-RFA-12-02 Postdoctoral Fellowships
Start Date:	12/01/2012	**End Date:**	11/30/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hargens, Alan (MENTOR/ University of California at San Diego)		
Grant/Contract No.:	NCC 9-58-PF03001		
Performance Goal No.:			
Performance Goal Text:			
	POSTDOCTORAL FELLOWSHIP These experiments will determine intracranial pressure (ICP), intraocular pressure (IOP), and ocular structures-function using state-of-the-art noninvasive techniques during conditions know to simulate microgravity induced head-ward fluid shifts in healthy astronaut-aged individuals. We hypothesize that a head-ward fluid shift during microgravity exposure alters translaminar pressure (ICP-IOP) at the optic nerve head, compromising ocular structures-function. We will test proven head-ward fluid shift countermeasures (lower body negative pressure exercise and Braslet) to mitigate elevations of ICP and IOP. The same device used for pre-, in- and post-flight ICP measures of astronauts will be utilized. These studies will employ a whole body tilt protocol known to induce head-ward fluid shifts and raise ICP, enabling direct comparisons of IOP, ICP, and ocular structures-function. Moreover, these studies will determine the efficacy of lower		

Task Description:	body negative pressure or Braslet countermeasures to mitigate elevations in ICP and IOP during conditions known to induce head-ward fluid shifts. Briefly, measures of ocular structure will include: optic nerve head and macula tomography, retinal nerve fiber layer thickness, anterior-to-posterior axial length of eyeball, and retrolaminar subarachnoidal space of cerebrospinal fluid (CSF). Measures of ocular function will include: best corrected visual acuity, total retinal blood flow, and capillary blood flows in the optic nerve head and in the macula area. The NASA Countermeasure Readiness Level (CRL) may be regarded 2 at the beginning of the study and the level will rise to 6 at the conclusion of the study if the countermeasure of lower body negative pressure or Braslet is successful. These experiments will establish standard tests, methodologies, and databases for the assessment of acute changes in ICP and IOP. These low-cost experiments will deliver rapid data that can be used as a foundation for future flight-analog and flight studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These studies will define how simulated microgravity affects fluid redistribution in brain and eye. Understanding how pressures of the brain and eye interact during simulated spaceflight will support human space exploration and enhance life on Earth of common clinical eye and brain pathologies through advances in space biomedical science and technology. We are proposing a well-documented and validated battery of non-invasive or minimally-invasive, image-based tests developed to identify and quantify microgravity-induced, head-ward volume and pressure changes (type and magnitude) and a prioritization of these changes as to their deleterious effects on vision.
Task Progress:	Over 30% of American astronauts after long duration space flight on the International Space Station (ISS) have developed structural changes to the eye that affect vision. In addition, some astronauts have elevated fluid pressures in the spine and head after ISS missions. The space flight environment is complex and several factors may contribute to these observed vision alterations. However, the movement of fluid towards the head is one of the primary factors believed to initiate vision changes during space flight. The ground-based experiments will determine intracranial pressure (ICP), intraocular pressure (IOP), and ocular structures-function using state-of-the-art noninvasive techniques during conditions know to simulate microgravity induced head-ward fluid shifts in healthy astronaut-aged individuals. We hypothesize that a head-ward fluid shift during microgravity exposure causes a change in translaminar pressures (ICP-IOP) at the optic nerve head, compromising ocular structures-function. We will test proven head-ward fluid shift countermeasures (e.g., lower body negative pressure (LBNP) and Braslet) to mitigate elevations of ICP and IOP. The same device used for pre-, in- and post-flight ICP measures of astronauts will be utilized. These studies will employ a whole body tilt protocol known to induce head-ward fluid shifts and raise ICP, enabling direct comparisons of IOP, ICP and ocular structures-function. Moreover, these studies will determine the efficacy of lower body negative pressure or Braslet countermeasures to mitigate elevations in ICP and IOP during conditions known to induce head-ward fluid shifts. During year-1 we obtained IRB approval, setup experimental hardware, and enrolled subjects to determine how posture and low body negative pressure (LBNP) affect intraocular pressure. Building on this proven Tilt and LBNP protocol, we have initiated setup for future cerebral and cochlear fluid pressure (CCFP) measurements, noninvasive ICP. In addition, we plan to compare LBNP and Braslet during these proven tilt conditions.
Bibliography Type:	Description: (Last Updated: 06/03/2026)
Abstracts for Journals and Proceedings	Boudreaux RD, Elmer K, Metzger C, Macias B, Allen M, Braby L, Hogan H, Bloomfield SA. "Radiation Exposure Prevents Recovery of Cancellous Bone in Mouse Vertebrae Following Partial Weightbearing." 35th Annual Meeting of the American Society for Bone and Mineral Research, Baltimore, Maryland, October 4-7, 2013. J Bone Miner Res 2013;28(Suppl 1):S389. Available at: http://www.asbmr.org/asbmr-2013-abstract-detail?aid=f29b33be-00b0-47b9-87fb-8af8b382b9df ; accessed 1/6/2014. , Oct-2013
Abstracts for Journals and Proceedings	Macias BR, Hargens AR, Liu JHK. "Lower Body Negative Pressure Counters Short-Duration Head-Down-Tilt Induced Elevation in Intraocular Pressure." 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. Program and Abstracts, p. 108. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Abstracts for Journals and Proceedings	Macias BR, Liu JHK, Hargens AR. "Altered Ocular Structure/Function and Mitigation of Peri-optic Nerve Edema during Simulated Microgravity." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013
Abstracts for Journals and Proceedings	Mao CP, Macias B, Hargens AR. "Skin Blood Flow and Muscle Oxygenation in the Shoulder during Backpack Carriage." Southwest American College of Sports Medicine Meeting, Newport Beach, CA, October 18-19, 2013. Southwest American College of Sports Medicine Meeting, Newport Beach, CA, October 18-19, 2013. , Oct-2013
Abstracts for Journals and Proceedings	Thao S, Macias BR, Hargens AR. "Leg Blood Flow and Limb Circumference with Altered External Pressure for Prosthesis in Sports." Southwest American College of Sports Medicine Meeting, Newport Beach, CA, October 18-19, 2013. Southwest American College of Sports Medicine Meeting, Newport Beach, CA, October 18-19, 2013. , Oct-2013
Articles in Peer-reviewed Journals	Swift JM, Lima F, Macias BR, Allen MR, Greene ES, Shirazi-Fard Y, Kupke JS, Hogan HA, Bloomfield SA. "Partial weight bearing does not prevent musculoskeletal losses associated with disuse." Med Sci Sports Exerc. 2013 Nov;45(11):2052-60. http://dx.doi.org/10.1249/MSS.0b013e318299c614 ; PubMed PMID: 23657172 , Nov-2013
Awards	Macias BR. "Distinguished Honor Graduate in Department of Health and Kinesiology, December 2013." Dec-2013